

Pairwise Sequence Alignment (2)

BioInf 2025/2026 1st Semester

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Summary

- Local Alignment
- Hybrid/Semi-global alignment
- Affine gap penalties
- Exercise

Global Alignment

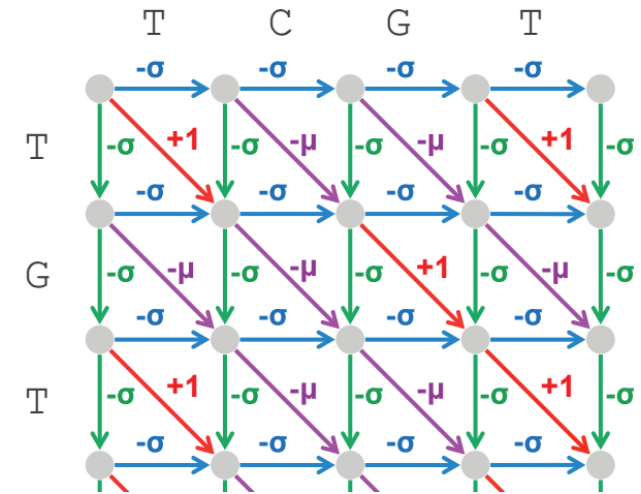
Global Alignment Problem:

Find a highest-scoring alignment of two strings as defined by a scoring matrix.

Input: Two strings and a scoring matrix *Score*.

Output: An alignment of the strings whose alignment score (as defined by *Score*) is maximized among all possible alignments of the strings.

- We use the Needleman-Wunch to find the longest path in the alignment graph



Global Alignment

- Sometimes a global alignment does not have biological significance
- If we want to find specific regions of concentrated matches
 - We want to focus on smaller regions and ignore the borders
- Global: 22 matches, 18 indels, and 2 mismatches

GCC–**C**–A**GTC**–**TATGT**–**CAG**GGGG**CACG**––**A**–**GCA****TGCACA**–
GCCGCC–**GTCGT**–**T**–**T****CAG**–––**CA**–**GTTATGT**–**T**–**CAGAT**

- Local: 17 matches and 32 indels

–––**G**––––**C**–––––**C**––**CAGTCTATG**–**TCAG**GGGG**CACGAGCA****TGCACA**
GCC**GCCGT****CGTTTT****CAGCAGT**–**TATGTTCAG**–––––**A**–––––**T**–––––

Sequence Alignment

- Sequence alignment for DNA (or in rarer cases RNA) and Proteins
 - Protein sequence alignment is an important tool for sequence annotation, i.e. providing functions to (parts of) sequences.
 - DNA sequence alignment, apart from annotation, is also used in other applications related with phylogenetic analysis.
- Two broad classes of sequence alignment:
 - **global** alignment: aligning the complete sequences.
 - **local** alignment: find good alignments of adjacent parts of both sequences, ignoring the remaining.

	Protein	DNA
Global	<pre>L G P S S G C A S R I W T K S A T G P S - G - - S - I W S K S G</pre>	<pre>- C A G T G C A T G - A C A T A T C A G - G C - T C T A C A G A</pre>
Local	<pre>L G P S S G C A S R I W T K S A T W N R - G C A S R I W M R D W</pre>	<pre>- C A G T G C A T G T A C A G A T T C G - T C - T G T A C A G T</pre>

Local Alignment

Local Alignment Problem:

Find the highest-scoring local alignment between two strings.

Input: Strings v and w as well as a scoring matrix $Score$.

Output: Substrings of v and w whose global alignment score (as defined by $Score$) is maximized among all substrings of v and w .

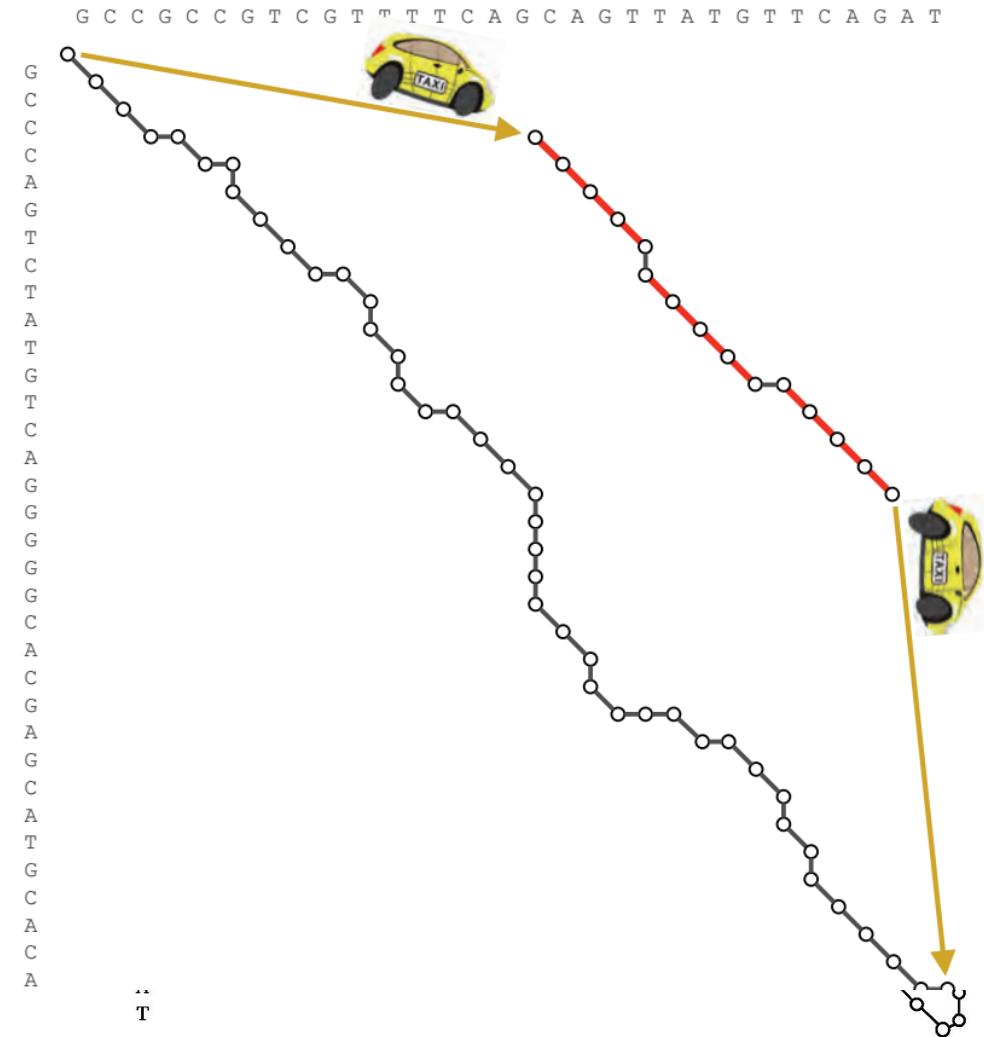
- We need to find longest path connecting any two nodes
 - Not very efficient
- What if we could jump right to the best start and end nodes?

Local Alignment

If we used this scoring function:

$$s_{i,j} = \max \begin{cases} 0 \\ s_{i-1,j} + \text{Score}(v_i, -) \\ s_{i,j-1} + \text{Score}(-, w_j) \\ s_{i-1,j-1} + \text{Score}(v_i, w_j) \end{cases}$$

We could do something like this:



Local Alignment

- Find good alignments of contiguous parts of both sequences, ignoring the remaining.
- We will not consider negative scores in the matrix S
 - only interested in the best matching subsequences,
- Tracing back the alignment: start with the highest score on the whole matrix and stop as soon as we find a 0.
- We are only aligning the best matching segments, so the remainder of the sequences are left out from the alignment.

Local Alignment - Smith-Waterman Algorithm (BA5F)

- The best alignment to A_{i-1}, B_{j-1} plus the match A_i, B_j
- Or the best alignment to A_i, B_{j-1} and align B_j with gap
- Or the best alignment to A_{i-1}, B_j and align A_i with gap
- Or restart (zero)

$$S_{i,j} = \max \begin{cases} S_{i-1,j-1} + sm(A_i, B_j) \\ S_{i,j-1} + g \\ S_{i-1,j} + g \\ 0 \end{cases}$$

Local Alignment - Smith-Waterman Algorithm

- $$S_{i,j} = \max \begin{cases} S_{i-1,j-1} + sm(A_i, B_j) \\ S_{i,j-1} + g \\ S_{i-1,j} + g \\ 0 \end{cases}$$
- Trace back from best value to first 0
- Only the best scoring stretch is represented; the rest is not aligned
- Example:

T A A T G A
 T A A - G A

	-	C	T	A	A	T	G	A	C	C	G
-	0	0	0	0	0	0	0	0	0	0	0
T	0	0	1	0	0	1	0	0	0	0	0
T	0	0	1	0	0	1	0	0	0	0	0
A	0	0	0	2	1	0	0	1	0	0	0
A	0	0	0	1	3	1	0	1	0	0	0
G	0	0	0	0	1	2	2	0	0	0	1
A	0	0	0	1	1	0	1	3	1	0	0

Local Alignment - Smith-Waterman Algorithm

- Trace back from best value to first 0
 - The trace back matrix T have four possible values:
 - the three as before (diagonal, horizontal, and vertical) and
 - an extra hypothesis corresponding to the case where the alignment is terminated (where the S matrix is filled with 0).
- Example:

	gap	H	G	W	A	G
gap	0	0	0	0	0	0
P	0	0	0	0	0	0
H	0	8	0	0	0	0
S	0	0	8	0	1	0
W	0	0	0	19	11	3
G	0	0	6	11	19	17

	gap	H	G	W	A	G
gap	0	0	0	0	0	0
P	0	0	0	0	0	0
H	0	8	0	0	0	0
S	0	0	8	0	1	0
W	0	0	0	19	11	3
G	0	0	6	11	19	17

Best alignments:

H	S	W
H	G	W

H	S	W	G
H	G	W	A

Semi-Global Alignment (Hybrid)

- Initialize the matrix with a value of 0:
 - first **row** to ignore gaps placed on the **left (column)** sequence.
 - first **column** to ignore gaps placed on the **top (row)** sequence,
- Sequence is free to “slide” and start at any point of the other sequence without penalties
- Gaps at the end of a sequence:
 - instead of tracing back from the bottom right cell:
 - Trace back from the highest scoring cell in the **last row** if we want to ignore gaps in the **left (column)** sequence,
 - or from the highest scoring cell in **the last column** if we want to ignore the gaps in **the top (row)** sequence.

Semi-Global Alignment

- Example: No start or end penalties on left (AGCT)
 - No penalties at start of AGCT because of 0 on top row.
 - No penalties at end of AGCT because we start from highest score in last row.

C C G C T A
- A G C T -

	-	C	C	G	C	T	A
-	0	0	0	0	0	0	0
A	-2	-1	-1	-1	-1	-1	1
G	-4	-3	-2	0	-2	-2	-1
C	-6	-3	-2	-2	1	-1	-3
T	-8	-5	-4	-3	-1	2	0

Semi-Global Alignment

- Example: No start penalties on left (AGCT) and no end penalties on top (AAAC)
 - No penalties at start of AGCT because of 0 on top row.
 - No penalties at end of AAAC because we start from highest score in last column.

A A A C -
- A G C T

	-	A	A	A	C
-	0	0	0	0	0
A	-2	1	1	1	-1
G	-4	-1	0	0	0
C	-6	-3	-2	-1	1
T	-8	-5	-4	-3	-1

Semi-Global Alignment

- Example: No end penalties on left (AGCT) and no start penalties on top (CTAA)
 - No penalties at end of AGCT because we start from highest score in last row.
 - No penalties at start of CTAA because of 0 in first column.

- - C T A A
A G C T - -

	-	C	T	A	A
-	0	-2	-4	-6	-8
A	0	-1	-3	-3	-5
G	0	-1	-2	-4	-4
C	0	1	-1	-3	-5
T	0	-1	2	0	-2

Semi-Global Alignment

- No start penalties:
 - On (first) left sequence: set first row to zero, to slide on top sequence
 - On (second) top sequence: set first column to zero, to slide on left sequence
- No end penalties:
 - On (first) left sequence: start from best of last row (best spot on top sequence)
 - On (second) top sequence: start from best on last column (best spot on left sequence)
- End gaps are not scored but they still appear on the alignment

Affine Gap Penalties (BA5J)

- Simplest alternative: use a single parameter (g) that defines a constant penalty for each column where a gap occurs.
 - protein sequence: -7 to -12
 - DNA: -2 to -3
- Affine gap penalty model: penalize heavily the start of a gap (gap opening penalty g), but do not penalize so much a gap extension (typically be of $r = -1$ or -2).
 - This complicates alignment algorithms a bit, with extra bookkeeping

Back to Non-Ribosomal Peptides

- With the correctly aligned sequences we can identify the positions that code for certain aminoacids

YAFDLGYTCMFPVLLGGGELHIVQKETYTAPDEIAHYIKEHGITYIKLTPSLFHTIVNTASFAFDANFESLRLIVLGGEKIIPIDVIAFRKMYGHTE-FINHYGPTTEATIGA
-AFDVSAGDFARALLTGGQLIVCPNEVKMDPASLYAIIKKYDITIFEATPALVIPLMEYI-YEQKLDISQLQILIVGSDSCSMEDFKTLVSRFGSTIRIVNSYGVTEACIDS
IAFDASSWEIYAPLLNGGTVVCIDYYTTIDIKALEAVFKQHHIRGAMLPALLKQCLVSA----PTMISSLEILFAAGDRLSSQDAILARRAVGSGV-Y-NAYGPTENTVLS

- If it was not correctly aligned, we would have this:

YAFDLGYTCMFPVLLGGGELHIVQKETYTAPDEIAHYIKEHGITYIKLTPSLFHTIVNTASFAFDANFESLRLIVLGGEKIIPIDVIAFRKMYGHTEFINHYGPTTEATIGA
AFDVSAGDFARALLTGGQLIVCPNEVKMDPASLYAIIKKYDITIFEATPALVIPLMEYIYEQKLDISQLQILIVGSDSCSMEDFKTLVSRFGSTIRIVNSYGVTEACIDS
IAFDASSWEIYAPLLNGGTVVCIDYYTTIDIKALEAVFKQHHIRGAMLPALLKQCLVSAPTMISSLEILFAAGDRLSSQDAILARRAVGSGVYNAYGPTENTVLS

Back to Non-Ribosomal Peptides

LTKVGHIG → Asp

VGEIGSID → Orn

AWMFAAVL → Val

- The purple positions were identified by studying the 3D structure of the protein
- We know the sequence of about 160 of those modules
- The non-ribosomal code is still not fully understood

Alignment practice

- Write the global alignment matrix for TGT with TCGT, assuming a gap penalty of -2
- Backtrack to obtain the full alignment

Summary

- Local Sequence Alignment (Smith-Waterman Algorithm)
- Semi-global Alignment
- Affine Gap Penalty

Further reading

- Rocha and Ferreira Chapter 6
- Compeau and Pevzner Chapter 5